Internet of Things and Cloud Computing – Building Connected Ecosystems

CSA1510 – Cloud Computing and Big Data Analytics

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INTRODUCTION

- **Definition**

IoT

➢Internet of Things (IoT) is a network of physical devices connected to the internet.

➢These devices collect and exchange data automatically.

➢Examples include smart watches, smart bulbs, and smart home systems.

Cloud Computing

➢Cloud Computing provides computing services over the internet.

➢It stores data and processes information without requiring local servers.

➢Popular cloud platforms include AWS, Microsoft Azure, and Google Cloud.

Core Technical Explanation

➢How IoT and Cloud Work Together

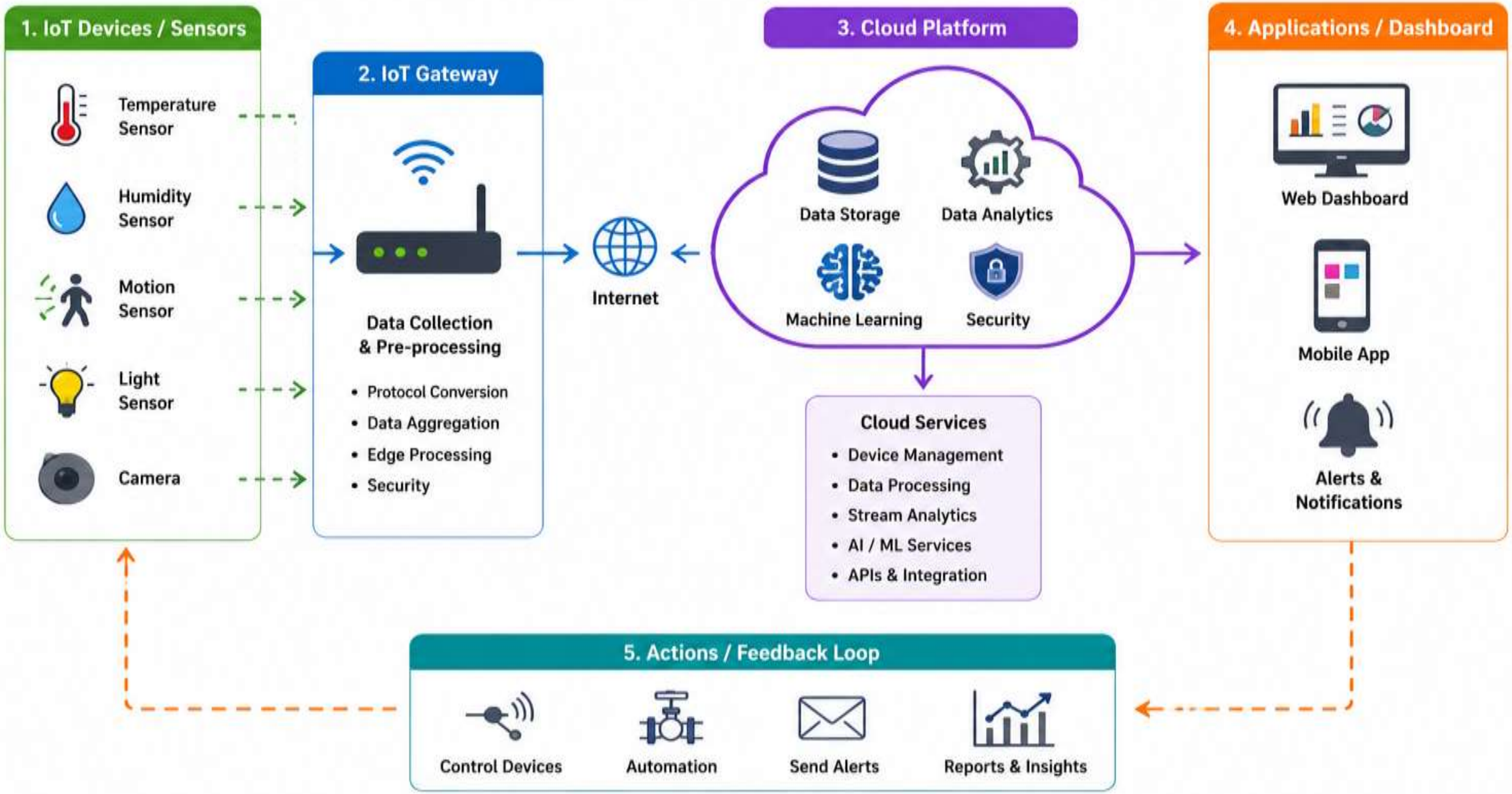
- IoT devices collect data using sensors.
- The collected data is sent through the internet.
- Cloud servers receive, store, and process the data.
- Processed information is sent back to users through mobile or web applications.
- Users can monitor and control devices from anywhere.

➢Benefits

- Real-time monitoring
- Remote access
- Large-scale data storage
- Fast data processing
- Better decision making

IoT and Cloud Architecture

Building Connected Ecosystems



Practical Example – Smart Agriculture

➢ Working Process

- Soil moisture sensors measure water levels.
- Temperature and humidity sensors collect environmental data.
- Information is sent to the cloud.
- Cloud analyses crop conditions.
- Farmers receive recommendations through a mobile app.
- Irrigation systems automatically turn on when needed.

➢ Benefits

- Saves water
- Increases crop production
- Reduces labor
- Provides real-time monitoring
- Supports precision farming

Industry Relevance

- **Applications of IoT and Cloud**
  - Smart Homes
  - Healthcare Monitoring
  - Smart Agriculture
  - Manufacturing Automation
  - Smart Cities
  - Transportation
  - Retail
  - Energy Management
- **Future Scope**
  - Integration with Artificial Intelligence (AI)
  - Growth of Edge Computing
  - Expansion of Smart Cities
  - Improved Security
  - Healthcare Innovations
  - Industrial Automation (Industry 4.0)

CONCLUSION

➢Conclusion

- IoT devices generate real-time data.
- Cloud computing stores, processes, and analyzes IoT data.
- Together, they build smart and connected ecosystems.
- They improve automation, efficiency, and decision-making across industries.
- IoT and Cloud Computing are transforming the future of technology.

➢Key Takeaway

- **IoT collects data, Cloud processes data, and together they enable intelligent, connected, and automated systems**

➢Advantages

- Automation
- Reduced operational cost
- Increased efficiency
- Better decision making
- Improved customer experience

THANK YOU